SOWETO GLOBAL SOLUTIONS — MASTER LEARNING SERIES

Computer Literacy to Advanced Excel

Complete Master Curriculum & Practical Analytics Reference Guide

Document Type: Master Curriculum Handbook

Target Audience: General Readers, Analysts & Professionals

Skill Range: Level 1 (Foundational) to Level 4 (Advanced)

Edition: 2026 Consolidated Edition

CURRICULUM OVERVIEW

This master curriculum provides a structured, progressive learning pathway designed to transition individuals from basic computer operating system usage to advanced spreadsheet data modeling, automated ETL processes, and executive dashboard design.

LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 4
Computer Literacy	**Intermediate Excel**	**Advanced Analytics**	**Data Modeling**

TABLE OF CONTENTS

Module / Level	Core Topic Coverage	Page
Level 1: OS & Systems	Hardware architecture, operating systems, and file navigation	Page 2
Level 1: System Admin	File permissions, security hygiene, and maintenance workflows	Page 3
Level 2: Excel Basics	Grid navigation, formatting standards, and structural best practices	Page 4
Level 2: Essential Formulas	Mathematical functions, logical IF statements, and referencing	Page 5
Level 3: Advanced Functions	Lookup formulas (XLOOKUP, INDEX/MATCH) and nesting	Page 6
Level 3: Summarization	Dynamic PivotTables, Slicers, and multi-dimensional grouping	Page 7
Level 3: Data Cleanliness	Data validation, deduplication, and conditional formatting	Page 8
Level 4: Power Query	Extract, Transform, Load (ETL) and automated imports	Page 9
Level 4: Data Modeling	Relational Data Models, Power Pivot, and DAX calculations	Page 10
Level 4: Dashboards	Executive dashboard construction and KPI design principles	Page 11
Mastery Checklist	Competency evaluation, sign-off matrix, and study milestones	Page 12

LEVEL 1: HARDWARE & SYSTEM ARCHITECTURE

Before mastering application software, professionals must understand physical computer components, data processing loops, and primary system metrics.

1.1 Physical Hardware Components

Component	Primary Function	Measuring Metric
CPU	Executes instructions and performs logical operations.	Gigahertz (GHz) / Cores
RAM	Volatile temporary working memory for active programs.	Gigabytes (GB) / MHz
SSD / HDD	Non-volatile permanent storage for files and OS.	Terabytes (TB) / MB/s Read
Motherboard	Main circuit board connecting CPU, memory, and expansion slots.	Form Factor (ATX)
GPU	Processes graphical rendering and visual computing workloads.	VRAM (GB)

1.2 Peripheral Classification

Input Devices

Hardware used to supply data and control signals to a system.

Examples: Keyboard, Optical Mouse, Document Scanner, Microphone, Barcode Reader.

Output Devices

Hardware used to convey results of data processing to human users.

Examples: LED Display Monitor, Laser Printer, Audio Speakers, Projectors.

1.3 Core Shortcuts for Efficiency

`Ctrl + C` **py:** Duplicates selected text, file, or object into system clipboard.

`Ctrl + V` **ste:** Inserts contents of system clipboard into target location.

`Ctrl + Z` **do:** Reverts the most recent action or entry.

`Alt + Tab` **Switcher:** Quickly toggles between open application windows.

`Win + E` **e Manager:** Directly opens System File Explorer.

LEVEL 1: FILE SYSTEMS & SECURITY HYGIENE

Efficient file organization and sound digital security practices form the essential foundation of modern digital literacy.

2.1 File Directory Conventions & Hierarchy

A well-organized directory prevents data redundancy, speeds up document retrieval, and simplifies enterprise backup processes.

> **Standard Corporate Directory Layout**
>
> `C:\CompanyData\Projects\2026\Q1_Financials\Report_2026-03-31.xlsx`
>
> Always use ISO-style date formats (YYYY-MM-DD) in file naming to maintain chronological sort order automatically.

2.2 Standard Common File Formats

Category	Extension	Description & Standard Software
Documents	.docx / .pdf	Formatted Microsoft Word text documents / Portable Document Format.
Data Tables	.xlsx / .csv	Excel Workbook / Plain text Comma-Separated Values dataset.
Graphics	.png / .svg	Lossless raster images / Scalable vector web graphics.
Compressed	.zip / .rar	Archived containers holding multiple compressed files.

2.3 Maintenance & Cybersecurity Workflow

STEP 1 **plement the 3-2-1 Backup Strategy:** Maintain at least 3 total copies of sensitive business data, across 2 different storage media types, with 1 copy stored in an offsite or cloud location.

STEP 2 **rage Optimization:** Periodically run system disk cleanup tools to eliminate temporary cache files, old downloads, and free up persistent drive space.

STEP 3 **ntity Protection:** Enforce Multi-Factor Authentication (MFA) across all enterprise user logins. MFA requires a primary password combined with a temporary security code.

LEVEL 2: INTERMEDIATE EXCEL & GRID FUNDAMENTALS

Spreadsheet software processes tabular data, runs calculations, and visually presents numeric analyses. Level 2 establishes core grid operational competency.

3.1 The Excel Interface & Terminology

Workbook vs. Worksheet

A **Workbook** is the primary file container (.xlsx). A **Worksheet** (or sheet) is an individual grid tab containing rows and columns.

Cells & Ranges

A **Cell** is an individual intersection (e.g., B4). A **Range** is a rectangular block of contiguous cells (e.g., A1:C10).

3.2 Number Formatting Best Practices

Storing numbers with proper data types is required for formulas to calculate accurately without returning errors.

Format Type	Visual Example	Usage Context
General	10450.5	Unformatted default numbers or raw values.
Currency / Accounting	$10,450.50	Monetary entries; aligned alignment for financial statements.
Percentage	12.5%	Ratios and growth figures (stored internally as 0.125).
Short Date	2026-10-05	Standard serial date representations for time series analysis.

3.3 Creating Official Excel Tables

Converting plain ranges into official Excel Tables (`Ctrl + T`) unlocks automated formatting, structured references, and dynamic recalculations.

1. Highlight the continuous dataset including row headers.
2. Press `Ctrl + T` or navigate to *Insert > Table*.
3. Confirm "My table has headers" and apply a structured layout style.

LEVEL 2: ESSENTIAL FORMULAS & LOGIC

Formulas execute automated operations on cell values. All formulas begin with an equal sign (`=`).

4.1 Essential Mathematical Functions

Function	Formula Syntax	Practical Description
SUM	`=SUM(A1:A50)`	Calculates total sum of all values in specified range.
AVERAGE	`=AVERAGE(B1:B50)`	Returns arithmetic mean of numeric range values.
COUNT	`=COUNT(C1:C50)`	Counts cells containing numeric entries only.
COUNTA	`=COUNTA(C1:C50)`	Counts non-empty cells (both text and numbers).

4.2 Relative vs. Absolute Cell Addressing

Understanding the dollar sign operator (`$`) is critical when copying formulas across rows and columns.

Relative Reference (A1)

Changes position relatively when copied to another cell. Used for row-by-row operations.

Absolute Reference (A1)

Locks both row and column position. Stays fixed on a specific input cell (e.g., tax rate).

4.3 Conditional Logic with IF Statement

The `IF` function tests a condition and yields distinct outcomes for True or False evaluations.

IF Formula Syntax & Structure

```
=IF(D2 >= 50, "Pass", "Fail")
```

Evaluates if cell D2 is greater than or equal to 50. If true, outputs "Pass"; otherwise, outputs "Fail".

LEVEL 3: ADVANCED LOOKUPS & RELATIONAL LINKING

Level 3 focuses on cross-referencing multiple data lists and dynamically searching tabular datasets.

5.1 Modern Lookup Solution: XLOOKUP

`XLOOKUP` replaces older lookup utilities. It can search both left and right without breaking when columns are altered.

> **XLOOKUP Full Syntax**
>
> `=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found])`

Parameter	Purpose	Practical Example
lookup_value	Item key to search for.	`E2` (Employee ID)
lookup_array	Column containing key values.	`A2:A500` (ID Column)
return_array	Column containing target data.	`C2:C500` (Salary Column)
if_not_found	Custom output if key is missing.	`"Not Found"`

5.2 Classic Flexible Alternative: INDEX & MATCH

For compatibility across legacy spreadsheet environments, combining `INDEX` and `MATCH` offers robust dynamic search capabilities.

> **INDEX / MATCH Formula Combination**
>
> `=INDEX(C2:C100, MATCH(E2, A2:A100, 0))`
>
> `MATCH` locates the specific row position of E2 inside range A2:A100; `INDEX` returns the corresponding cell value from column C.

5.3 Multi-Condition Logic Functions

SUMIFS `UMIFS(Sum_Range, Criteria_Range1, Criteria1, ...)` — Sums values matching multiple simultaneous conditions.

COUNTIFS `NTIFS(Criteria_Range1, Criteria1, ...)` — Counts total rows meeting all specified criteria.

LEVEL 3: DATA SUMMARIZATION & PIVOTTABLES

PivotTables allow rapid, interactive aggregation of complex tabular datasets without requiring custom code or manual formulas.

6.1 PivotTable Architecture Elements

Rows Drop Zone

Categorical fields placed here group summary data into distinct vertical row headers.

Columns Drop Zone

Fields placed here spread categories horizontally (ideal for dates, quarters, or years).

Values Drop Zone

Numeric metrics summarized via SUM, AVERAGE, COUNT, or DISTINCT COUNT operations.

Filters / Slicers

Global controls allowing interactive filtering of entire summary reports with one click.

6.2 Step-by-Step PivotTable Workflow

STEP 1 **ect Clean Raw Source Data:** Confirm source dataset contains continuous rows, distinct single-row headers, and zero merged cells.

STEP 2 **eate Pivot Structure:** Navigate to *Insert > PivotTable* and select placement on a new worksheet tab.

STEP 3 **g & Drop Fields:** Place *Region* into Rows, *Product Category* into Columns, and *Sales Revenue* into Values.

STEP 4 **rmat Value Fields:** Right-click numeric values, open *Number Format*, and apply currency formatting standards.

LEVEL 3: DATA HYGIENE & FORMATTING RULES

Inaccurate, duplicate, or unvalidated data degrades operational analytics. Maintaining structured data hygiene rules is vital.

7.1 Data Validation Setup

Data Validation restricts user inputs on specific input cells to pre-approved lists or exact numeric ranges.

1. Highlight target user entry cells.
2. Navigate to *Data* > *Data Validation*.
3. Select *List* under Validation Criteria Allow settings.
4. Enter comma-separated values or link to a lookup range (e.g., `Approved, Pending, Rejected`).

7.2 Conditional Formatting Application

Conditional Formatting automatically highlights metrics based on specified visual threshold rules.

Rule Goal	Highlight Type	Visual Result
Highlight Overdue Invoices	Soft Red Fill	Dates past current system date.
Top 10% Revenue Performers	Green Text & Fill	Top numeric percentile entries.
Data Bars	In-cell Progress Bar	Visual comparison of magnitude across cells.

7.3 Removing Duplicates

To eliminate duplicate entries, select your table, navigate to *Data* > *Remove Duplicates*, select key identifier columns (e.g., Transaction ID), and confirm removal.

LEVEL 4: ADVANCED ETL WITH POWER QUERY

Level 4 introduces automated data engineering. Power Query automates data extraction, transformation, and loading (ETL) without writing code.

8.1 Power Query Processing Pipeline

STEP 1	STEP 2	STEP 3
Connect Source	**Transform Data**	**Load Model**

8.2 Essential Power Query Transformations

Transformation Action	Operation Purpose
Unpivot Columns	Converts horizontal matrix reports into clean 2D relational data tables.
Split Column by Delimiter	Splits combined fields (e.g., "Last, First") into distinct attribute columns.
Group By	Aggregates detailed transaction rows into summarized categorical metrics.
Merge Queries	Executes relational database joins (e.g., Left Outer Join) between two tables.

8.3 Automated Data Refresh Workflow

> **The Power of Query Automation**
>
> Once a Power Query pipeline is designed, updating the dataset requires simply clicking **Data > Refresh All**. The engine re-applies all cleansing steps automatically.

LEVEL 4: DATA MODELING & POWER PIVOT

Power Pivot transitions workbooks into multi-table relational data models capable of holding millions of records smoothly.

9.1 Star Schema Data Modeling Standards

Connect Fact tables (transactions) to Dimension tables (lookup categories) using primary key relationships.

Fact Tables

Contains high-volume numerical measurements, metrics, and transaction logs (e.g., Sales Orders).

Dimension Tables

Contains descriptive lookup attributes used for filtering (e.g., Customers, Products, Date Table).

9.2 Fundamentals of DAX Expressions

Data Analysis Expressions (DAX) is the functional modeling language used inside Power Pivot data models.

DAX Measure Example	Functional Mechanism
`Total Revenue := SUM(Sales[Amount])`	Calculates basic explicit sum over a table column.
`YTD Sales := TOTALYTD([Total Revenue], 'Calendar'[Date])`	Calculates year-to-date running sales total using a continuous calendar table.

LEVEL 4: EXECUTIVE DASHBOARD ARCHITECTURE

Dashboards summarize complex analytical models into clear, interactive executive visual presentations.

10.1 Four-Layer Workbook Layout Standard

Layer 1 **w Data Layer:** Hidden tabs containing raw inputs or Power Query pipeline connections.

Layer 2 **ta Model Layer:** Relational Power Pivot backend handling core DAX calculations.

Layer 3 **culation Engine Tab:** Hidden calculation sheet running dynamic array staging functions.

Layer 4 **esentation Dashboard Tab:** Clean, locked summary view displaying KPI Cards and PivotCharts.

10.2 KPI Card & Chart Design Best Practices

Visual Element	Design Standard
KPI Stat Callout	Large bold metric (e.g., Total Revenue) paired with a small trend indicator (+5.2%).
Line Chart	Reserved for displaying continuous metrics over uniform time intervals.
Bar / Column Chart	Ideal for comparing discrete categorical items side by side.

MASTERY CHECKLIST & SIGN-OFF MATRIX

Evaluate your practical skill progression across all four curriculum capability tiers before requesting formal assessment.

Capability Evaluation Checklist

- [ ] **Level 1 Mastered:** Hardware concepts, system directory structures, shortcuts, and 3-2-1 backup hygiene.
- [ ] **Level 2 Mastered:** Table creation (`Ctrl+T`), cell formatting, `SUM` , `AVERAGE` , absolute referencing (`$`), and basic `IF` logic.
- [ ] **Level 3 Mastered:** Writing `XLOOKUP` , building dynamic PivotTables with Slicers, and applying Data Validation rules.
- [ ] **Level 4 Mastered:** Automating data cleaning via Power Query, designing Star Schema models, and constructing interactive executive dashboards.

Formal Competency Sign-Off

Learner Name: _____ **Evaluator Name:** _____

Learner Signature: _____ **Evaluator Signature:** _____

Date of Completion: _____